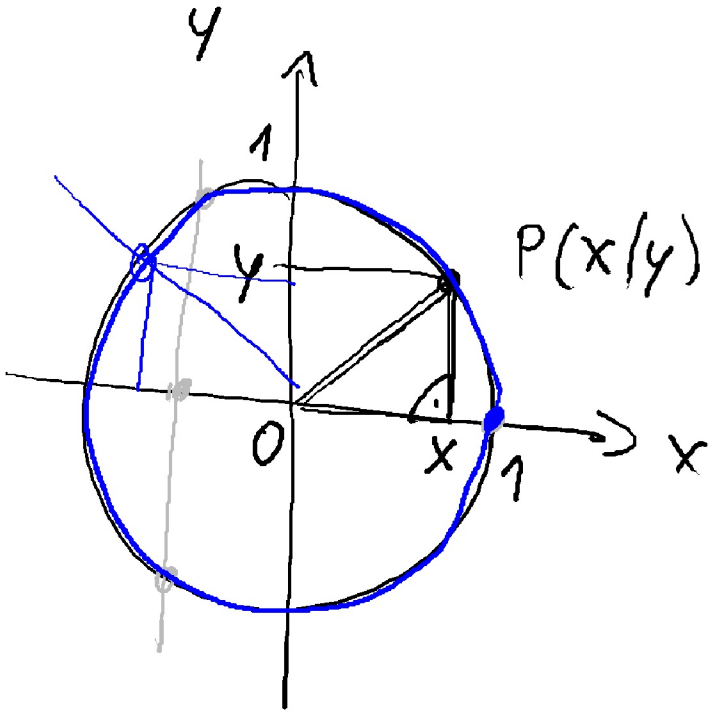


$$x^2 + y^2 = 1^2$$

$$y^2 = 1 - x^2$$

$$y = \pm\sqrt{1 - x^2}$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \frac{x^6}{6!} + \dots$$

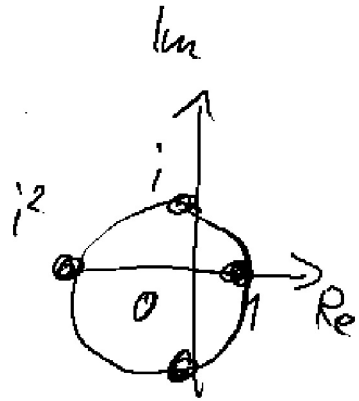
$$e^{ix} = 1 + ix + \frac{i^2x^2}{2!} + \frac{i^3x^3}{3!} + \frac{i^4x^4}{4!} + \frac{i^5x^5}{5!} + \frac{i^6x^6}{6!} + \dots$$

$$= \underline{1} + \underline{ix} - \underline{\frac{x^2}{2!}} - \underline{i\frac{x^3}{3!}} + \underline{\frac{x^4}{4!}} + i\frac{x^5}{5!} - \frac{x^6}{6!} + \dots$$

$$= \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots\right)$$

$$+ i\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} \dots\right)$$

$$= \cos x + i \sin x$$



$$\cos x = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$$

$$\sin x = \frac{1}{2i}\left(e^{ix} - e^{-ix}\right)$$

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$(\cos x)' = -\sin x$$

$$(\cosh x)' = \left(\frac{1}{2}(e^{x}+e^{-x})\right)'$$

$$= \frac{1}{2}\left(e^{x}-e^{-x}\right) = \sinh x$$

$$(\tanh x)' = \left(\frac{\sinh x}{\cosh x}\right)' = \frac{\cosh x \cdot \cosh x - \sinh x \cdot \sinh x}{\cosh^2 x}$$

$$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x}$$

$$(\operatorname{arsinh} x)' = \frac{1}{\cosh(\operatorname{arsinh} x)}$$

$$f(x) = \sinh x$$

$$f^{-1}(x) = \operatorname{arsinh} x$$

$$f'(x) = \cosh x$$

$$(f^{-1}(x))' = \frac{1}{f'(f^{-1}(x))}$$

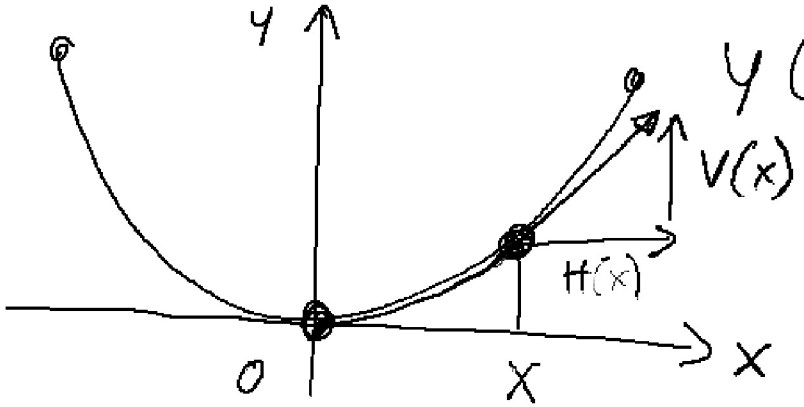
$$\cosh^2 x - \sinh^2 x = 1$$

$$\cosh^2 x = 1 + \sinh^2 x$$

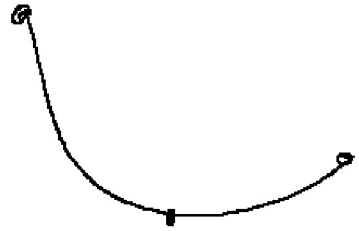
$$\cosh x = \pm\sqrt{1 + \sinh^2 x}$$

$$(\operatorname{arsinh} x)' = \frac{1}{\pm\sqrt{1 + \sinh^2(\operatorname{arsinh} x)}} = \pm\frac{1}{\sqrt{1 + x^2}}$$

$$= \frac{1}{\sqrt{1+x^2}}$$



$y(x) \longrightarrow$ berechnen



1. $H(x) = H_0 = \text{const}$

2. $V(x) = \mu \cdot s(x) \cdot g$

μ: Masse/Länge
$s(x)$: Länge

$$y'(x) = \frac{V(x)}{H(x)} = \frac{\mu g \cdot s(x)}{H_0}$$

$$= \underbrace{\frac{\mu g}{H_0}}_{k = \text{const.}} \cdot s(x)$$

$$y'(x) = k \cdot s(x) \qquad s(x) = \int_0^x \sqrt{1+(y'(t))^2}\,dt$$

$$y''(x) = k \cdot \sqrt{1+(y'(x))^2} \qquad s'(x) = \sqrt{1+y'(x)^2}$$

$$z(x) = y'(x)$$

$$z'(x) = y''(x)$$

$$z'(x) = k\sqrt{1+(z(x))^2}$$

$$\frac{dz}{dx} = z' = k\sqrt{1+z^2}$$

$$\int \frac{dz}{\sqrt{1+z^2}} = \int k\,dx$$

$$\operatorname{arsinh} z = kx + C_1$$

$$y' = z = \sinh(kx + C_1)$$

$$y = \cosh(kx + C_1) + C_2$$